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328813 (28)

BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Et & T

INDUSTRIAL & POWER ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Assume suitable data wherever necessary. Part (a) is compulsory in all questions and carries 2 marks. Solve any two parts out of (b), (c) and (d) in each question of 7 marks each.

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Unit-I

Q. 1. (a) Compare linear power supply and SMPS. 2

(b) Describe the circuit operation of transistorized series voltage regulator with advantage and disadvantage. 7

(c) Explain principle of operation of SMPS with the help of block diagram. 7

(d) Design a 10V zener regulator for the following conditions : 7

(i) The load current ranges from 50 mA to 100 mA.

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(ii) The source (input) voltage ranges from 12 V to 15 V.

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Unit-II

Q. 2. (a) Why Germanium is not used in SCR ? 2

(b) Draw and explain two transistor analogy of the SCR and describe the condition for its conduction ? 7

(c) Describe the various turning ON methods of SCR. Justify, SCR is a current operated device. 7

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(d) Classify thyristor commutation technique and explain them in brief. 7

Unit-III

Q. 3. (a) Give two comparison between 180° mode and 120° mode three-phase bridge inverter. 2

(b) Explain circulating current mode of single phase dual converter with advantage and disadvantage. 7

(c) Explain the principle of DC chopper circuit. What purpose do that serve ? 7

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(d) Explain mid-point configuration of single-phase cycloconverter ? 7

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Unit-IV

Q. 4. (a) What is skin effect. 2

(b) Explain principle of induction heating. Enlist merits of induction heating over conventional methods. 7

(c) Explain the basic principle of high frequency dielectric heating. Derive expression for heating power per unit volume. Give two applications. 7

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(d) Explain resistance welding. Explain the operation of sequence timer used in resistance welding control. 7

Unit-V

Q. 5. (a) What are the effect of noise on computer and communication systems. 2

(b) What are the different forms of Noise? Explain in brief. 7

(c) Explain the principle of buck-boost control of servo controlled voltage stablizer. 7

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(d) Draw block diagram of UPS and explain it. 7

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